

Nickname:

1.2 Classwork: Sampling and Histograms

1. There are 420 students at a school. The principal wants to know what the students think of the new lunch menu. She stands at the cafeteria door one Tuesday and asks the first 30 students who walk in:

“Do you agree that the new lunch menu is a big improvement?”

- (a) Write down the population, and write down the size of the sample.
- (b) The principal took the students who happened to be closest to hand. Name this sampling technique.
- (c) Give one reason why her sample may not be representative of all 420 students.
- (d) The question she asked is a *leading* question. Explain what is wrong with it, and rewrite it so that it is not leading.
- (e) Describe how the principal could instead take a simple random sample of 30 students.

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2. The height of each of 24 tomato plants was measured, in centimetres:

23.4	31.7	28.0	36.2	41.5	27.3
29.8	25.0	39.1	30.0	22.6	33.4
37.9	29.5	26.8	43.2	35.0	31.1
24.7	32.6	28.9	40.3	26.1	42.0

- (a) Complete the grouped frequency table below.
- (b) Write down the modal class.
- (c) One plant measured exactly 30.0 cm. State which class it is counted in, and explain why.
- (d) Draw a histogram for the data on the grid below.

Height h (cm)	Tally	Frequency
$20 \leq h < 25$		
$25 \leq h < 30$		
$30 \leq h < 35$		
$35 \leq h < 40$		
$40 \leq h < 45$		
Total		

